



BESS

ARENQ®
Forward To Future

Arenq Series

Liquid Cooling PV & ESS All-in-One System AQ125KW261KWH-P



BESS

The AQ125KW261KWH-P liquid-cooled energy storage system is a highly integrated solution combining LFP batteries, a composite power control unit (BMS, PCS, EMS), a fire suppression system, a thermal management system, and a cloud-based monitoring system.

Featuring an innovative integrated design, this system delivers superior safety standards, simplified management, and maximized profits. It is designed to flexibly meet diverse energy demands, providing a low-carbon, high-yield solution for various C&I application scenarios.



All-in-One Integration & Multi-Scenario Adaptability

- Adopts an all-in-one design, enabling seamless compatibility with PV, grid, and diesel genset integration to support diverse application scenarios.
- Fully integrates all core components: battery modules, PCS, STS, BMS, EMS, fire suppression



Intelligent Energy Management & Efficiency Optimization

- Enables advanced energy management, supporting functions such as demand control and anti-backfeed.
- Provides dynamic reactive power compensation (DRPC) for immediate power factor optimization, enhancing energy utilization efficiency.



Comprehensive Safety Protection

- Incorporates critical safety features, including black-start capability and robust protection against overcurrent and power backflow.
- Equipped with seamless grid-bypass functionality to directly supply utility power to the backup port, ensuring continuous operation.



High-Yield & Low-Carbon Solution

- Delivers superior safety standards, simplified management, and maximized profits through its innovative integrated design.
- Offers a low-carbon, high-yield solution tailored for various commercial and industrial (C&I) application scenarios.



Liquid Cooling PV & ESS All-in-One System

AQ125KW261KWH-P



Technical Parameter

System	
Nominal Capacity	314 Ah
Nominal Energy	261 kWh
Usable Energy	≥ 243 kWh
Rated Output Power	125 kW
Ingress Protection	1P54
Operating Temperature	-20°C ~ 55°C
Cooling Method	Liquid Cooling
Fire Suppression	Aerosol
Altitude	4000 m (>2000 m Derating)
Cycle Life	~ 8000 Cycles
Dimensions (W × D × H)	1400 × 1350 × 2250 mm
Weight	~ 2100 kg
Certifications	IEC 62619 / CE/UN38.3
Output AC (Grid side)	
Rated output power	125 kW
Max. apparent output power	125 KVA
Rated grid voltage	3/N/PE, 220V/380 V 3/N/PE, 230V/400 V
Rated grid frequency	50HZ/60Hz
Rated grid output current	189.9A/180.4A
Power Factor	-0.99 ~ +0.99
THDi	<3%
Output AC (Back-up)	
Rated output power	125kW
Max. apparent output power	1.2 times of rated power, 10 mins 1.4 times of rated power, 100s 1.6 times of rated power, 10s 2 times of rated power, 200ms
Back-up switch time	<10ms
Rated output voltage	3/N/PE, 220V/380 V 3/N/PE, 230V/400 V
Rated frequency	50 Hz/60 Hz
THDv(@linear load)	<2%
PV Side	
Max. PV Input Power	250 kW
MPPT Voltage Range	150 V - 950 V
Max. DC Input Current	10 × 42 A
No. of MPPTs / Max. Strings	10 / 20
Liquid Cooling System	
Cooling Capacity	5 KW
Rated Circulating Flow Rate	46.5 L/min @ 60 kPa
Refrigerant	R134a
Coolant Type	50% Ethylene Glycol Solution